

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	11.1 / 1.32381
Time	2026-07-02 02:00 UTC
Location	30.0395, -95.6739
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	4 / 371	43.99 to 100; mean 73.4325	70.16 at 2026-07-02 01:55Z; dt 5 min
asos	temperature	degC	4 / 371	22.7778 to 34.4444; mean 29.1171	29 at 2026-07-02 01:55Z; dt 5 min
asos	wind_direction	degree	4 / 562	0 to 360; mean 120	150 at 2026-07-02 01:55Z; dt 5 min
asos	wind_speed	m/s	4 / 575	0 to 11.8322; mean 2.9981	7.20222 at 2026-07-02 01:55Z; dt 5 min
noaa_gfs	gh_500	m	5 / 60	5894.32 to 5929.29; mean 5912.17	5902.41 at 2026-07-02 00:00Z; dt 120 min
noaa_gfs	pbl_height	m	5 / 60	34.8903 to 2463.44; mean 840.125	207.647 at 2026-07-02 00:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	5 / 60	37.1521 to 56.5395; mean 45.5085	50.81 at 2026-07-02 00:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	5 / 60	1006.37 to 1013.78; mean 1011.07	1006.37 at 2026-07-02 00:00Z; dt 120 min
noaa_gfs	t_850	degC	5 / 60	17.459 to 19.5649; mean 18.6861	19.2318 at 2026-07-02 00:00Z; dt 120 min
noaa_gfs	u_10m	m/s	5 / 60	-4.80877 to 4.01123; mean -0.8825	-0.395796 at 2026-07-02 00:00Z; dt 120 min
noaa_gfs	v_10m	m/s	5 / 60	-2.79096 to 5.8039; mean 2.1411	1.69125 at 2026-07-02 00:00Z; dt 120 min
openaq	ozone	ppm	7 / 454	0 to 0.065; mean 0.0192	0.008 at 2026-07-02 02:00Z; dt 0 min
openaq	pm10	ug/m3	1 / 73	22 to 114; mean 47.6575	64 at 2026-07-02 02:00Z; dt 0 min
openaq	pm25	ug/m3	60 / 2992	0 to 27.8; mean 5.4732	6.5 at 2026-07-02 02:00Z; dt 0 min
openweather	cloud_cover	percent	3 / 217	0 to 100; mean 24.0553	21 at 2026-07-02 02:00Z; dt 0.6 min
openweather	humidity	percent	3 / 217	53 to 93; mean 74.765	71 at 2026-07-02 02:00Z; dt 0.6 min
openweather	precipitation	mm/h	3 / 217	0 to 0; mean 0	0 at 2026-07-02 02:00Z; dt 0.6 min
openweather	pressure	hPa	3 / 217	1013 to 1017; mean 1015.4	1014 at 2026-07-02 02:00Z; dt 0.6 min
openweather	temperature	degC	3 / 217	24.58 to 35.33; mean 29.9199	30.51 at 2026-07-02 02:00Z; dt 0.6 min
openweather	wind_direction	degree	3 / 217	0 to 344; mean 148.931	153 at 2026-07-02 02:00Z; dt 0.6 min
openweather	wind_speed	m/s	3 / 217	0.45 to 5.36; mean 1.8361	1.79 at 2026-07-02 02:00Z; dt 0.6 min
purpleair	pm25	ug/m3	42 / 2947	0 to 97.1; mean 6.8087	6.2 at 2026-07-02 01:59Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_co_column	mol/m^2	8 / 8	0.0261592 to 0.0295937; mean 0.0276	0.0269498 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_hcho_column	mol/m^2	8 / 8	0.000150572 to 0.000285154; mean 0.0002	0.000228552 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	3.73882e-05 to 6.16311e-05; mean 0.0001	6.14335e-05 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_so2_column	mol/m^2	8 / 8	-0.000214267 to 2.39858e-05; mean -0	-1.38712e-05 at 2026-07-01 19:07Z; dt 412.4 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; dt 412.4 min
tceq	co	ppm	1 / 65	0.1 to 0.9; mean 0.3954	0.3 at 2026-07-02 02:00Z; dt 0 min
tceq	no2	ppb	6 / 375	0.5 to 26.9; mean 5.2293	11.1 at 2026-07-02 02:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-30 12Z 63.77, 18Z 55.27, 07-01 00Z 77, 06Z 91.6, 12Z 68, 18Z 54.58, 07-02 00Z 76.49, 06Z 91.58, 12Z 70.12, 18Z 61.04, 07-03 00Z 78.34, 06Z 90.14, 12Z 80.16
asos	temperature	06-30 12Z 31.18, 18Z 32.87, 07-01 00Z 28.58, 06Z 26.1, 12Z 30.59, 18Z 32.76, 07-02 00Z 27.98, 06Z 25.66, 12Z 30.01, 18Z 31.18, 07-03 00Z 27.85, 06Z 25.24, 12Z 28.28
asos	wind_direction	06-30 12Z 160, 18Z 148.5, 07-01 00Z 146.2, 06Z 118.8, 12Z 143.6, 18Z 149.1, 07-02 00Z 130.2, 06Z 71.88, 12Z 105, 18Z 124.8, 07-03 00Z 124.2, 06Z 54.68, 12Z 39.29
asos	wind_speed	06-30 12Z 4.099, 18Z 5.562, 07-01 00Z 4.534, 06Z 1.972, 12Z 2.54, 18Z 4.341, 07-02 00Z 3.172, 06Z 1.072, 12Z 1.64, 18Z 4.619, 07-03 00Z 2.84, 06Z 0.6645, 12Z 0.7202
noaa_gfs	gh_500	06-30 18Z 5929, 07-01 00Z 5920, 06Z 5922, 12Z 5907, 18Z 5919, 07-02 00Z 5902, 06Z 5907, 12Z 5895, 18Z 5908, 07-03 00Z 5904, 06Z 5919, 12Z 5914
noaa_gfs	pbl_height	06-30 18Z 2345, 07-01 00Z 1134, 06Z 619.9, 12Z 142.1, 18Z 2192, 07-02 00Z 959.9, 06Z 304.6, 12Z 124.7, 18Z 1917, 07-03 00Z 185.6, 06Z 88.17, 12Z 68.94
noaa_gfs	precipitable_water	06-30 18Z 43.81, 07-01 00Z 43.12, 06Z 39.41, 12Z 38.27, 18Z 41.36, 07-02 00Z 47.82, 06Z 47.11, 12Z 49.31, 18Z 52.57, 07-03 00Z 54.6, 06Z 46.35, 12Z 42.37
noaa_gfs	surface_pressure	06-30 18Z 1012, 07-01 00Z 1010, 06Z 1011, 12Z 1011, 18Z 1011, 07-02 00Z 1009, 06Z 1011, 12Z 1011, 18Z 1012, 07-03 00Z 1011, 06Z 1012, 12Z 1012
noaa_gfs	t_850	06-30 18Z 18.46, 07-01 00Z 18.77, 06Z 19.47, 12Z 18.78, 18Z 18.4, 07-02 00Z 19.2, 06Z 18.85, 12Z 18.37, 18Z 17.62, 07-03 00Z 18.68, 06Z 18.86, 12Z 18.77
noaa_gfs	u_10m	06-30 18Z -1.067, 07-01 00Z -3.054, 06Z -1.626, 12Z -0.2487, 18Z -0.8201, 07-02 00Z -2.446, 06Z 0.07146, 12Z 0.5124, 18Z -1.019, 07-03 00Z -0.4448, 06Z -0.4377, 12Z -0.01089

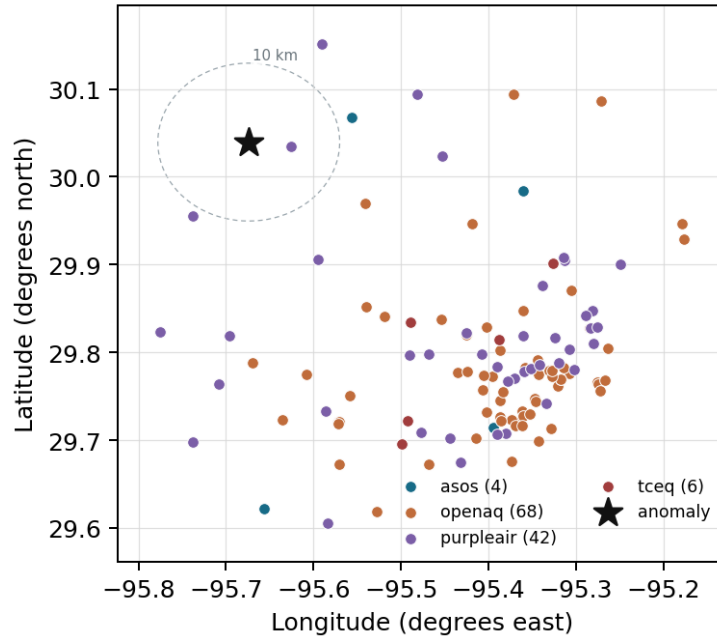
Source	Metric	Values
noaa_gfs	v_10m	06-30 18Z 3.396, 07-01 00Z 4.51, 06Z 3.156, 12Z 1.054, 18Z 2.52, 07-02 00Z 3.077, 06Z 2.343, 12Z 1.328, 18Z 1.638, 07-03 00Z -0.141, 06Z 1.746, 12Z 1.065
openaq	ozone	06-30 12Z 0.01908, 18Z 0.02863, 07-01 00Z 0.01689, 06Z 0.01221, 12Z 0.01695, 18Z 0.03511, 07-02 00Z 0.01779, 06Z 0.005743, 12Z 0.01317, 18Z 0.03788, 07-03 00Z 0.02129, 06Z 0.006543, 12Z 0.006619
openaq	pm10	06-30 12Z 46.75, 18Z 39.67, 07-01 00Z 34.17, 06Z 53.33, 12Z 50.5, 18Z 52.83, 07-02 00Z 60.33, 06Z 48.67, 12Z 70.33, 18Z 37.33, 07-03 00Z 36, 06Z 44.67, 12Z 41.67
openaq	pm25	06-30 12Z 5.546, 18Z 3.828, 07-01 00Z 5.468, 06Z 6.117, 12Z 6.551, 18Z 5.183, 07-02 00Z 5.292, 06Z 6.316, 12Z 6.113, 18Z 3.019, 07-03 00Z 4.564, 06Z 5.76, 12Z 6.357
openweather	cloud_cover	06-30 12Z 1.846, 18Z 3.833, 07-01 00Z 0.4118, 06Z 0.8947, 12Z 2.333, 18Z 8.389, 07-02 00Z 19.35, 06Z 37.74, 12Z 68.83, 18Z 85.06, 07-03 00Z 55.39, 06Z 4.778, 12Z 1.833
openweather	humidity	06-30 12Z 71.85, 18Z 61.17, 07-01 00Z 75.24, 06Z 87.47, 12Z 75.22, 18Z 57.61, 07-02 00Z 71.88, 06Z 86.79, 12Z 76.33, 18Z 63.11, 07-03 00Z 77.39, 06Z 86.39, 12Z 88
openweather	precipitation	06-30 12Z 0, 18Z 0, 07-01 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-02 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-03 00Z 0, 06Z 0, 12Z 0
openweather	pressure	06-30 12Z 1017, 18Z 1015, 07-01 00Z 1015, 06Z 1015, 12Z 1016, 18Z 1014, 07-02 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1015, 07-03 00Z 1015, 06Z 1016, 12Z 1017
openweather	temperature	06-30 12Z 31.11, 18Z 33.98, 07-01 00Z 29.73, 06Z 26.91, 12Z 30.41, 18Z 34.61, 07-02 00Z 29.89, 06Z 26.48, 12Z 29.96, 18Z 33.06, 07-03 00Z 28.56, 06Z 26.06, 12Z 26.68
openweather	wind_direction	06-30 12Z 139.5, 18Z 144.6, 07-01 00Z 123.1, 06Z 132.6, 12Z 169.9, 18Z 153.8, 07-02 00Z 143.7, 06Z 169.6, 12Z 139.3, 18Z 129.3, 07-03 00Z 165.1, 06Z 165.9, 12Z 167.5
openweather	wind_speed	06-30 12Z 1.961, 18Z 2.955, 07-01 00Z 2.103, 06Z 1.386, 12Z 1.566, 18Z 2.484, 07-02 00Z 1.657, 06Z 1.428, 12Z 1.157, 18Z 2.648, 07-03 00Z 1.612, 06Z 1.466, 12Z 0.9267
purpleair	pm25	06-30 12Z 6.121, 18Z 4.907, 07-01 00Z 6.241, 06Z 7.424, 12Z 7.42, 18Z 5.826, 07-02 00Z 7.076, 06Z 8.757, 12Z 8.11, 18Z 4.793, 07-03 00Z 5.8, 06Z 8.188, 12Z 9.935
sentinel5p	s5p_aer_ai_granule_available	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	s5p_ch4_granule_available	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	s5p_co_column	06-30 18Z 0.02626, 07-01 18Z 0.02689, 07-02 18Z 0.02856
sentinel5p	s5p_co_granule_available	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	s5p_hcho_column	06-30 18Z 0.0001509, 07-01 18Z 0.0002309, 07-02 18Z 0.0002773
sentinel5p	s5p_hcho_granule_available	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	s5p_no2_column	06-30 18Z 3.739e-05, 07-01 18Z 6.143e-05, 07-02 18Z 5.589e-05
sentinel5p	s5p_no2_granule_available	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	s5p_o3_granule_available	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	s5p_so2_column	06-30 18Z 1.808e-05, 07-01 18Z 1.938e-06, 07-02 18Z -8.53e-05
sentinel5p	s5p_so2_granule_available	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
tceq	co	06-30 12Z 0.35, 18Z 0.3167, 07-01 00Z 0.3667, 06Z 0.3667, 12Z 0.46, 18Z 0.3667, 07-02 00Z 0.375, 06Z 0.45, 12Z 0.4167, 18Z 0.4333, 07-03 00Z 0.425, 06Z 0.3333, 12Z 0.5333
tceq	no2	06-30 12Z 4.539, 18Z 3.256, 07-01 00Z 2.737, 06Z 5.064, 12Z 6.177, 18Z 4.642, 07-02 00Z 5.132, 06Z 5.586, 12Z 6.187, 18Z 6.755, 07-03 00Z 5.845, 06Z 5.728, 12Z 5.728

Per-site averages

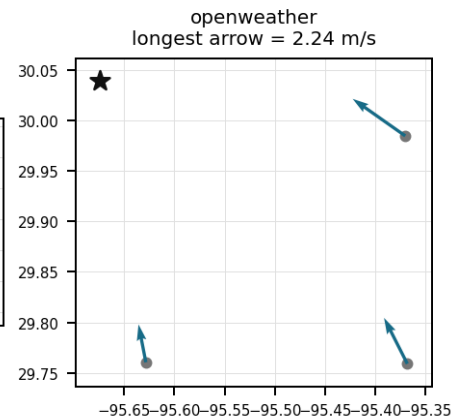
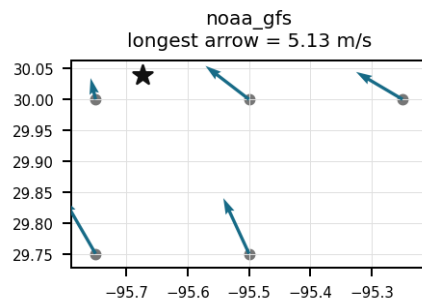
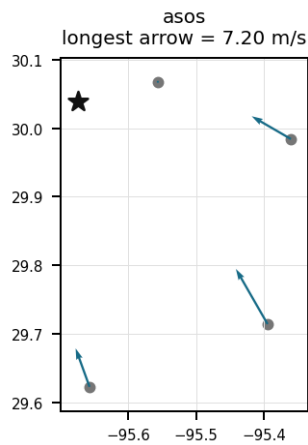
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	DWH (11.767 km) 76.69, IAH (30.778 km) 72.53, MCJ (45.095 km) 71.1, SGR (46.424 km) 75.37
asos	temperature	DWH (11.767 km) 28.09, IAH (30.778 km) 29.44, MCJ (45.095 km) 29.55, SGR (46.424 km) 29.06
asos	wind_direction	DWH (11.767 km) 83.61, IAH (30.778 km) 126, MCJ (45.095 km) 137.3, SGR (46.424 km) 134.7
asos	wind_speed	DWH (11.767 km) 1.354, IAH (30.778 km) 3.283, MCJ (45.095 km) 4.098, SGR (46.424 km) 3.259
noaa_gfs	gh_500	gfs:30.00,-95.75 (8.54 km) 5912, gfs:30.00,-95.50 (17.314 km) 5912, gfs:29.75,-95.75 (33.018 km) 5912, gfs:29.75,-95.50 (36.299 km) 5912, gfs:30.00,-95.25 (41.053 km) 5913
noaa_gfs	pbl_height	gfs:30.00,-95.75 (8.54 km) 653.8, gfs:30.00,-95.50 (17.314 km) 1062, gfs:29.75,-95.75 (33.018 km) 696.4, gfs:29.75,-95.50 (36.299 km) 928.7, gfs:30.00,-95.25 (41.053 km) 859.9
noaa_gfs	precipitable_water	gfs:30.00,-95.75 (8.54 km) 45.37, gfs:30.00,-95.50 (17.314 km) 45.84, gfs:29.75,-95.75 (33.018 km) 45.26, gfs:29.75,-95.50 (36.299 km) 44.9, gfs:30.00,-95.25 (41.053 km) 46.17
noaa_gfs	surface_pressure	gfs:30.00,-95.75 (8.54 km) 1009, gfs:30.00,-95.50 (17.314 km) 1011, gfs:29.75,-95.75 (33.018 km) 1011, gfs:29.75,-95.50 (36.299 km) 1012, gfs:30.00,-95.25 (41.053 km) 1012
noaa_gfs	t_850	gfs:30.00,-95.75 (8.54 km) 18.66, gfs:30.00,-95.50 (17.314 km) 18.68, gfs:29.75,-95.75 (33.018 km) 18.69, gfs:29.75,-95.50 (36.299 km) 18.73, gfs:30.00,-95.25 (41.053 km) 18.67
noaa_gfs	u_10m	gfs:30.00,-95.75 (8.54 km) -0.5441, gfs:30.00,-95.50 (17.314 km) -0.8033, gfs:29.75,-95.75 (33.018 km) -0.9633, gfs:29.75,-95.50 (36.299 km) -0.8733, gfs:30.00,-95.25 (41.053 km) -1.228
noaa_gfs	v_10m	gfs:30.00,-95.75 (8.54 km) 2.53, gfs:30.00,-95.50 (17.314 km) 2.126, gfs:29.75,-95.75 (33.018 km) 2.315, gfs:29.75,-95.50 (36.299 km) 2.141, gfs:30.00,-95.25 (41.053 km) 1.595
openaq	ozone	3723 (0.015 km) 0.01989, 319 (28.951 km) 0.0182, 287 (35.354 km) 0.01817, 310 (36.873 km) 0.02229, 1319333 (39.972 km) 0.02147, 311 (41.77 km) 0.01713, 2046 (44.308 km) 0.01675
openaq	pm25	14140749 (15.005 km) 6.826, 14140750 (24.564 km) 6.753, 14157977 (26.633 km) 7.044, 10762809 (26.733 km) 7.129, 14140753 (27.9 km) 6.745, 14103803 (29.758 km) 4.301, 14152860 (30.129 km) 1.329, 12337668 (30.849 km) 6.902, +52 more
openweather	cloud_cover	grid:north (29.906 km) 25.15, grid:west (31.345 km) 23.81, grid:center (42.955 km) 23.22
openweather	humidity	grid:north (29.906 km) 75.35, grid:west (31.345 km) 74.29, grid:center (42.955 km) 74.66
openweather	precipitation	grid:north (29.906 km) 0, grid:west (31.345 km) 0, grid:center (42.955 km) 0
openweather	pressure	grid:north (29.906 km) 1015, grid:west (31.345 km) 1015, grid:center (42.955 km) 1015
openweather	temperature	grid:north (29.906 km) 29.72, grid:west (31.345 km) 29.78, grid:center (42.955 km) 30.26
openweather	wind_direction	grid:north (29.906 km) 128.1, grid:west (31.345 km) 157.4, grid:center (42.955 km) 161.2
openweather	wind_speed	grid:north (29.906 km) 1.992, grid:west (31.345 km) 1.598, grid:center (42.955 km) 1.917
purpleair	pm25	217323 (4.664 km) 7.654, 3033 (11.204 km) 5.182, 288282 (14.805 km) 9.782, 130763 (16.666 km) 2.203, 194469 (19.477 km) 10.2, 165203 (21.37 km) 5.454, 98139 (24.624 km) 7.854, 223797 (25.981 km) 6.417, +34 more
tceq	no2	482010029 (0.0 km) 3.157, 482010047 (28.955 km) 4.928, 482010024 (36.874 km) 4.33, 482011052 (37.243 km) 13.08, 482011066 (39.435 km) 2.821, 482010055 (41.774 km) 2.548

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 2 of 35

A fresh local combustion plume is supported because the TCEQ monitor at the anomaly site measured NO2 at 11.1 ppb while nearby ozone was only 0.008 ppm at the same time, a pairing consistent with recently emitted NO titrating ozone before substantial downwind aging.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 3 of 35

A broad regional pollution or smoke episode is contradicted because particulate matter remained moderate near the event, Sentinel-5P CO was not unusually elevated before the event, and Sentinel-5P SO2 showed no corresponding enhancement, which argues against a large-area transported combustion or sulfur-rich industrial plume as the dominant cause.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 4 of 35

A fresh nearby combustion emission plume, most plausibly from on-road traffic, diesel activity, industrial combustion, or power generation near the monitor, is a likely cause of the sharp NO₂ enhancement because the NO₂ spike occurred with simultaneously suppressed ozone, a pattern characteristic of recent NO titration in a young NO_x plume.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 5 of 35

Elevated regional tropospheric NO₂ column densities observed by Sentinel-5P satellite overpasses indicate urban-scale NO_x buildup across the Houston region prior to the surface event.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 6 of 35

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM_{2.5} and NO₂.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 7 of 35

Nocturnal boundary layer collapse and low surface wind speeds trapped local vehicular and industrial nitrogen oxide emissions near the ground, driving the surface NO₂ anomaly detected by TCEQ monitors.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 8 of 35

NOAA GFS modeled planetary boundary layer height dropped significantly overnight alongside low ASOS surface wind speeds, supporting nocturnal boundary layer stagnation during the July 2 NO₂ spike.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 9 of 35

Industrial activities and vehicle traffic are likely contributing factors to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 10 of 35

Regional-scale alternative explanations are less supported because Sentinel-5P showed only modest NO₂ enhancement on the prior daytime overpass, Sentinel-5P CO and SO₂ did not show a strong regional plume, and surface PM_{2.5} stayed near typical single-digit micrograms per cubic meter levels around the event.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 11 of 35

The PM2.5 levels are elevated, with a mean of 6.8087 ug/m3 and a range of 0.0 ug/m3 to 97.1 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 12 of 35

The meteorological environment near 2026-07-02T02:00:00+00:00 featured humid air and modest south-southeasterly flow, and model guidance indicated a much shallower boundary layer later that night, which supports temporary near-surface trapping and poor dispersion of a local NOx plume.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 13 of 35

The anomaly occurred on July 2nd, around 02:00:38+00:00, with the nearest sensor reading 6.2 ug/m3 at a distance of 43.489 km from the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 14 of 35

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and volatile organic compounds into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 15 of 35

Nighttime accumulation under a shallow and stabilizing boundary layer is a likely contributing cause of the elevated surface NO₂ because the event occurred near 02:00 UTC after sunset, with GFS boundary-layer height dropping from about 960 m at 00Z toward about 305 m by 06Z while humidity was high, conditions that commonly reduce vertical dilution of primary NO_x.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 16 of 35

The air quality anomaly is related to PM_{2.5} levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 17 of 35

The tceq NO₂ measurement at 2026-07-02T02:00:00+00:00 at 30.040, -95.674 was 11.1 ppb, compared with an expected value of 1.323809523809524 ppb, yielding a z-score of 6.726124255396563 and a severity classification of severe.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 18 of 35

OpenAQ surface ozone measurements dropped to near-zero nighttime levels near 0.0057 ppm at the time of the TCEQ NO2 peak, supporting chemical titration of ozone by local emissions.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 19 of 35

Tropospheric NO2 column concentrations over the study area reached a peak 6-hour mean of 6.143e-05 mol/m² on 2026-07-01 at 18:00 UTC prior to the ground-level event.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 20 of 35

The nearest openaq ozone measurement at the anomaly time and location was 0.008 ppm at 2026-07-02T02:00:00+00:00, 0.015 km from 30.040, -95.674.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 21 of 35

The TCEQ NO2 monitor at 30.040, -95.674 measured 11.1 ppb at 2026-07-02T02:00:00+00:00, far above its expected value of 1.323809523809524 ppb, while nearby ozone was only 0.008 ppm at the same time, which is consistent with a fresh local NOx plume undergoing ozone titration.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 22 of 35

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 23 of 35

The combination of stagnant air and high humidity is trapping pollutants near the surface, exacerbating the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 24 of 35

The anomaly is more consistent with a localized short-duration surface event than with a large regional smoke or photochemical episode because particulate matter and carbon monoxide were not simultaneously extreme, satellite SO₂ was not elevated, and the same-period ozone near the site was low rather than high.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 25 of 35

NOAA GFS model output showed a shallow nocturnal boundary layer height dropping below 310 meters during the early morning hours of July 2, 2026, suppressing vertical atmospheric mixing.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 26 of 35

Nighttime accumulation under reduced mixing is supported because the anomaly occurred near local evening with high humidity around 70 to 76 percent and generally modest local winds near 1.8 to 3.2 m/s, while GFS boundary-layer height decreased from about 960 m at 00Z toward about 305 m by 06Z, indicating a transition toward a shallower nocturnal layer that could temporarily elevate surface NO₂.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 27 of 35

Nocturnal titration of ambient ozone by fresh nitric oxide emissions converted NO into NO₂ under suppressed vertical mixing, reflected by near-zero O₃ levels recorded in OpenAQ ground data during the NO₂ peak.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 28 of 35

Sentinel-5P NO₂ total column density measurements peaked at 6.14e-5 mol/m² on July 1 before the surface event, providing evidence of regional tropospheric NO_x accumulation across the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 29 of 35

The combination of high PM_{2.5} and NO₂ levels is a significant concern for public health in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 30 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO₂ particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 31 of 35

A severe NO₂ anomaly with a concentration of 11.1 ppb, representing a z-score of 6.73 above the expected value of 1.32 ppb, was detected at station 482010029 (30.040, -95.674) on 2026-07-02 at 02:00:00 UTC.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 32 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 33 of 35

TCEQ ground monitoring detected a severe nitrogen dioxide spike reaching 11.1 ppb at 02:00 UTC on July 2, 2026, representing a z-score of 6.73 above expected values.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 34 of 35

The pollutant with the detected anomaly is nitrogen dioxide (NO₂) measured by tceq.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 35 of 35

Sentinel-5P satellite measurements recorded elevated tropospheric nitrogen dioxide column densities exceeding $5.5\text{e-}05 \text{ mol/m}^2$ over the Houston area during surrounding satellite passes.

Mark exactly one:☐ V☐ I☐ U**Note:**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
